

Learning and Efficiency in the Market for Physician Referrals

SUPPLEMENTAL APPENDIX PRELIMINARY

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1 Descriptive Statistics

This section provides additional details of our analytic data and subsequent referral patterns.

2 Identification

We formally show identification of the myopic model, and then we discuss identification of the dynamic model.

$$\alpha m_{ijt} + f(e_{ijt}) + u(x_{ijt}) + c(n_{jt}, z_j) + \xi_j + \epsilon_{ijt}$$

SLIDE

* Both Dickstein (2021) and Gong (2018) omit α , which fixes the MRS between patient outcomes and other factors (incl. ϵ) at an assumed value

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- Altruism parameter (α): identified by marginal effect of *observed* success rate, in limit as $e \rightarrow \infty$ (“ID at infinity”)

$$\alpha m_{ijt} = \alpha \frac{a_0 + \sum_{s=1}^{t-1} Y_{ijs}}{a_0 + b_0 + \sum_{s=1}^{t-1} D_{ijs}} = \alpha \frac{a_0/e_{ijt} + \bar{y}_{ijt}}{(a_0 + b_0)/e_{ijt} + 1}$$

where $\bar{y}_{ijt} \equiv \sum_{s=1}^{t-1} Y_{ijs} / \sum_{s=1}^{t-1} D_{ijs}$ (and $e_{ijt} \equiv \sum_{s=1}^{t-1} D_{ijs}$)

- Strength of priors ($a_0 + b_0$): how the marginal effect of the success rate changes with experience (i.e., interaction)

$$\alpha m_{ijt} = \alpha \frac{a_0}{(a_0 + b_0) + e_{ijt}} + \alpha \frac{e_{ijt}}{(a_0 + b_0) + e_{ijt}} \times \bar{y}_{ijt}$$

- Prior mean ($\frac{a_0}{a_0 + b_0}$): estimate outside the model, using average success probability in each market (then have a_0, b_0 separately)

2.1 Myopic Model

Given a distribution for the shocks ϵ , the conditional choice probabilities identify differences in utility between the alternatives. The effects of the pairwise characteristics, $u(x)$, and the specialist fixed effects, ξ , are identified subject to standard normalizations (e.g., one fixed effect is set equal to zero). Because the number of patients at a specialist, n_{jt} , is endogenous, the congestion effect, $c(n_{jt}, z_j)$, is identified with an instrument for the probability that patients *other than patient t* see specialist j . A natural instrument for this is the distances between *other* patients and specialist j (see Richards-Shubik et al., 2022).

This leaves the terms αm and $f(e)$. The parameter α is identified in the limit as the number of past patients sent to a specialist grows large (i.e., so-called “identification at infinity”). Recall $e_{ijt} \equiv \sum_{s=1}^{t-1} D_{ijs}$ and let $\bar{y}_{ijt} \equiv \sum_{s=1}^{t-1} Y_{ijs} / e_{ijt}$. Then,

$$\alpha m_{ijt} = \alpha \frac{a_0 + \sum_{s=1}^{t-1} Y_{ijs}}{a_0 + b_0 + \sum_{s=1}^{t-1} D_{ijs}} = \alpha \frac{a_0/e_{ijt} + \bar{y}_{ijt}}{(a_0 + b_0)/e_{ijt} + 1}.$$

Hence, $\lim_{e_{ijt} \rightarrow \infty} \alpha m_{ijt} = \alpha \bar{y}_{ijt}$. Intuitively, α is identified by variation in the success rate among specialists to whom a PCP has sent many patients in the past.

Next, the strength of the priors, $(a_0 + b_0)$, is identified by how the marginal effect of the success rate changes with the number of past patients. Rearranging αm_{ijt} differently, we have

$$\alpha m_{ijt} = \alpha \frac{a_0}{(a_0 + b_0) + e_{ijt}} + \alpha \frac{e_{ijt}}{(a_0 + b_0) + e_{ijt}} \times \bar{y}_{ijt}. \quad (1)$$

With α identified, the second term in equation (1) has only one unknown parameter, $\eta \equiv (a_0 + b_0)$. The marginal effect of $\bar{y}$ increases with e , and η governs that interaction.

To identify the prior mean, $\rho \equiv \frac{a_0}{a_0 + b_0}$, we use the observed average success rate in the market. This assumes that PCPs have a general awareness of the overall success rate of the procedure in their market. It might instead be possible to identify a_0 from the choice data, but this would require a strong functional form assumption to separate the term $\alpha \frac{a_0}{(a_0 + b_0) + e}$ from the familiarity effect $f(e)$.

Finally, then, the familiarity effect is identified by the remainder of the marginal effect of experience with a specialist that is not absorbed by αm (i.e., what is left over from $\alpha \frac{a_0}{(a_0 + b_0) + e}$). A standard normalization is required, such as $f(0) = 0$. Also f must be bounded as $e \rightarrow \infty$; otherwise, familiarity would dominate the effects of all other attributes as $e \rightarrow \infty$. This does not seem desirable from a modeling perspective, and it would nullify the “identification at infinity” argument for α .

We start by proving the identification of the myopic model. To review (and simplify), the utility to PCP i for referring patient t to specialist j is

$$U_{ijt} \equiv \alpha m_{ijt} + f(e_{ijt}) + \xi_j + \epsilon_{ijt},$$

where m_{ijt} is the current mean of the beliefs about the probability of success at specialist j (using the history before patient t), e_{ijt} is the amount of experience with specialist j (the number of past patients i sent to j), $f(e_{ijt})$ is the subjective utility from familiarity with specialist j , ξ_j is an unobservable specialist fixed effect, and ϵ_{ijt} is an idiosyncratic shock with a known distribution. Also let $\bar{y}_{ijt}$ denote the number of successes among past patients that i sent to j . NOTE: There is no outside option with utility normalized to zero.

Given a parametric assumption on the error term, ϵ , the observable choice probabilities directly identify differences in utility, excluding the error term: e.g., $(U_{ijt} - \epsilon_{ijt}) - (U_{ikt} - \epsilon_{ikt})$.

To simplify these expressions, let v denote utility without the error term: $v \equiv U - \epsilon$.

To be somewhat more precise, there is a finite number of specialists but large numbers of patients and PCPs. The choice probabilities are

$$\Pr(j | (e_{ikt}, \check{y}_{ikt})_{k=1}^J),$$

where “ j ” means the event that specialist j is chosen. We may need consistent estimates of the choice probabilities for cases where $e_{ikt} = 0, \forall k$, and where $e_{ikt} \rightarrow \infty, \forall k$. Hence the number of PCPs may need to go to infinity, but the number of patients must grow faster.

We identify the [other] components of v in steps as follows.

1. Specialist fixed effects (ξ): Any cases where the observables are the same for two specialists can be used to identify the fixed effects (up to a standard normalization; e.g., one fixed effect is set to zero). Because if $e, \check{y}, x, n$ are the same, then

$$v_{ijt} - v_{ikt} = \xi_j - \xi_k.$$

In particular, cases where $e = 0$ (and hence $\check{y} = 0$) are useful here.

2. Altruism parameter (α): This uses cases with large past experience, formally the limit when $e \rightarrow \infty$. Let $e_{ijt} \rightarrow \infty$ and $e_{ikt} \rightarrow \infty$ (and for convenience $x_{ijt} = x_{ikt}$ and $n_j = n_k$). Then

$$v_{ijt} - v_{ikt} = \frac{\alpha \bar{y}_{ijt}}{0+1} + \xi_j - \frac{\alpha \bar{y}_{ikt}}{0+1} - \xi_k.$$

[I CAN PULL UP MORE OF THE ALGEBRA FROM BELOW.] Hence

$$\alpha = \frac{(v_{ijt} - \xi_j) - (v_{ikt} - \xi_k)}{\bar{y}_{ijt} - \bar{y}_{ikt}}$$

in these limits. Thus cases where a PCP has large past experience with two specialists, but different success rates with them, identify the altruism parameter.

3. Strength of priors ($a_0 + b_0$): The responsiveness to success with finite experience then identifies the strength of priors (i.e., the stickiness of beliefs). Let $e_{ijt} = e_{ikt} = e$ (and

for convenience $x_{ijt} = x_{ikt}$ and $n_j = n_k$). Then

$$v_{ijt} - v_{ikt} = \frac{\alpha \check{y}_{ijt}}{a_0 + b_0 + e} + \xi_j - \frac{\alpha \check{y}_{ikt}}{a_0 + b_0 + e} - \xi_k.$$

Hence

$$(a_0 + b_0) = \frac{\alpha(\check{y}_{ijt} - \check{y}_{ikt})}{(v_{ijt} - \xi_j) - (v_{ikt} - \xi_k)} - e,$$

and all the objects on the right are observable or already identified.

4. Effect of familiarity ($f(e)$): Finally, differences in experience with specialists identify the effect of familiarity. For convenience let x, n be the same, then

$$v_{ijt} - v_{ikt} = \alpha \frac{a_0 + \check{y}_{ijt}}{a_0 + b_0 + e_{ijt}} + f(e_{ijt}) + \xi_j - \alpha \frac{a_0 + \check{y}_{ikt}}{a_0 + b_0 + e_{ikt}} - f(e_{ikt}) - \xi_k.$$

Hence

$$\begin{aligned} f(e_{ijt}) - f(e_{ikt}) &= (v_{ijt} - \xi_j) - (v_{ikt} - \xi_k) \\ &= \alpha \left[\frac{a_0}{a_0 + b_0 + e_{ijt}} - \frac{a_0}{a_0 + b_0 + e_{ikt}} \right] \\ &= \alpha \left[\frac{\check{y}_{ijt}}{a_0 + b_0 + e_{ijt}} - \frac{\check{y}_{ikt}}{a_0 + b_0 + e_{ikt}} \right]. \end{aligned}$$

This identifies f up to a normalization, specifically $f(0) = 0$.

However this also requires the value of a_0 , which can be obtained if the prior ratio, $\frac{a_0}{a_0 + b_0}$, is calibrated separately. Intuitively, we need to fully identify the priors (both the ratio and the strength) in order to recover the effect of familiarity, because we need to know exactly how experience washes away the priors in order to separate that from the effect of familiarity.

QED

2.2 Finite Sample Performance

We recover negative values of α in the majority of markets, whenever η is set above a relatively small number. This may occur because of the way that α attaches to both the

number of successes and the number of patients in the model, as follows:

$$\alpha \frac{\rho\eta + y_{ijt}}{\eta + e_{ijt}}.$$

This can be split into two terms where the coefficient is constrained to be the same (because ρ and η are set at fixed values):

$$\alpha \frac{\rho\eta}{\eta + e_{ijt}} + \alpha \frac{y_{ijt}}{\eta + e_{ijt}}.$$

There is relatively little variation in y conditional on e , because the success rate is high. Consequently, much of the identifying variation for α comes from the first term (driven by variation in e) not the second term (driven by variation in y conditional on e).

If PCPs are more likely to refer patients to specialists to whom they have sent more patients in the past, then e would have a positive relationship with the specified utility function. Hence the first term, with e in the denominator, would contribute to a negative estimate of α . The model includes a separate familiarity effect, specified as $\beta \log(1 + e)$, which should absorb this, but the distribution of the estimates of α across HRRs is similar whether we include that term or not.

3 Estimation

To constrain $\alpha \geq 0$ we estimate $\ln \alpha$ and specify the beliefs term as $\exp(\ln \alpha) \cdot m$. We use an iterative procedure proposed in which makes it possible to employ standard estimation commands designed for multinomial logits where all terms are linear in parameters. The procedure is based on a first-order Taylor series approximation of the term that is nonlinear in parameters. With the exponential function this is simply

$$\exp(\ln \alpha_2) \cdot m \approx [\exp(\ln \alpha_1) + (\ln \alpha_2 - \ln \alpha_1) \exp(\ln \alpha_1)] \cdot m,$$

(because $d \exp(\ln \alpha) / d \ln \alpha = \exp(\ln \alpha)$), where $\ln \alpha_1$ and $\ln \alpha_2$ are two candidate values.

In our case, the iteration works by setting a value of $\ln \alpha$ (e.g., $\ln \alpha_1$) and then estimating the model with a free coefficient (e.g., $\hat{\pi}_1$) on the term $\exp(\ln \alpha_1) \cdot m$. Rearranging the Taylor

approximation above shows how the estimated coefficient can be used to update the value of $\ln \alpha$, as follows:

$$\exp(\ln \alpha_2) \cdot m \approx \underbrace{[1 + (\ln \alpha_2 - \ln \alpha_1)]}_{\hat{\pi}_1} \exp(\ln \alpha_1) \cdot m;$$

hence:

$$\ln \alpha_2 = \ln \alpha_1 + \hat{\pi}_1 - 1.$$

Furthermore, because the estimated coefficient $\hat{\pi}_1$ by definition improves the likelihood, using $\exp(\ln \alpha_2)$ should produce a higher likelihood than $\exp(\ln \alpha_1)$. For some intuition note that at the maximum, $\ln \alpha_*$, the likelihood cannot be improved, and so the value of $\hat{\pi}$ should be one; i.e., in terms of the Taylor expansion

$$\exp(\ln \alpha_*) = \underbrace{[1 + (\ln \alpha_* - \ln \alpha_*)]}_{\hat{\pi}_*} \exp(\ln \alpha_*) = \underbrace{[1]}_{\hat{\pi}_*} \exp(\ln \alpha_*)$$

The standard error then follows as

$$SE(\hat{\alpha}) = \exp(\ln \alpha_*) SE(\hat{\pi}_*)$$

because $\hat{\alpha} = \exp(\ln \alpha_*) = \exp(\ln \alpha_* + \hat{\pi}_* - 1)$ and asymptotically we only need to consider the variation in $\hat{\pi}_*$.

4 Counterfactuals without Familiarity

The purpose of this counterfactual is to estimate the potential changes in referrals and subsequently patient health if PCPs could rely entirely on specialist quality, with full information, and without any attachment due to prior relationships. These results are presented in Figure 2. Here, we estimate larger effects on referral probabilities, with reallocation of 33.5 percent on average across HRRs. The health effects of such a scenario nonetheless remain relatively small, with an increase in the *ex ante* probability of a successful surgery of around 0.4 percentage points on average.

References

- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, “The Effect of Minimum Wages on Low-Wage Jobs,” *The Quarterly Journal of Economics*, August 2019, *134* (3), 1405–1454.
- Richards-Shubik, Seth, Mark S. Roberts, and Julie M. Donohue, “Measuring quality effects in equilibrium,” *Journal of Health Economics*, May 2022, *83*, 102616.

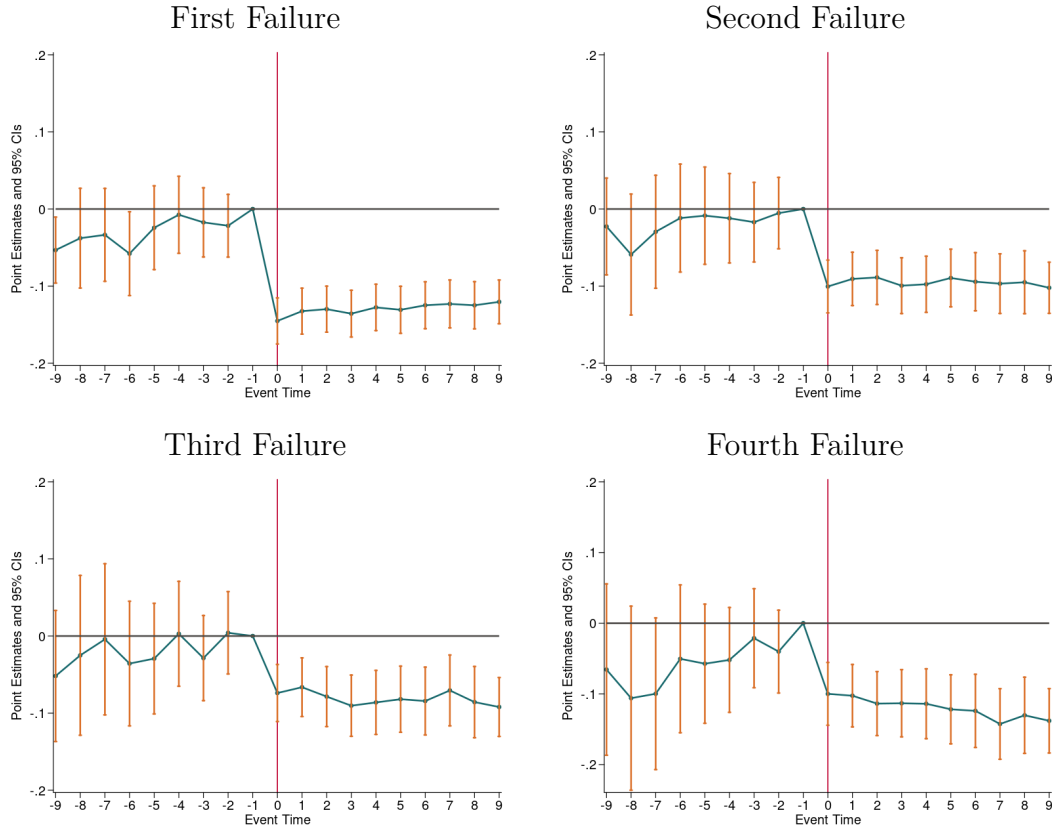
Tables and Figures

Table 1: **Sensitivity Analysis**^a

Parameter	Mean	Percentile				
		10th	25th	50th	75th	90th
Baseline	0.1924	0.0000	0.0000	0.1063	0.2766	0.5499
Integrated	0.1921	0.0000	0.0000	0.1122	0.2785	0.5015
Peer Outcomes	0.2033	0.0000	0.0000	0.1137	0.3132	0.5811

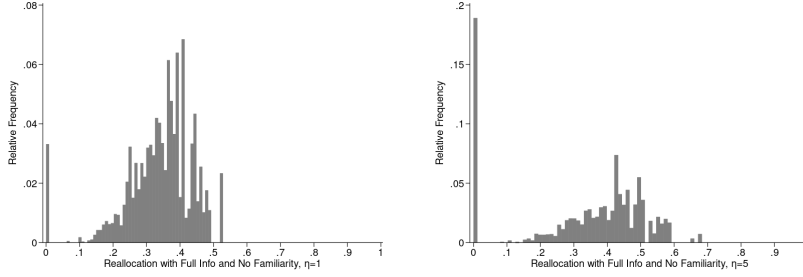
^aEstimates of α (for $\eta = 1$) across alternative specifications. “Baseline” denotes our original estimates for the myopic learning model presented in the main text; “Integrated” denotes the estimates for α when including a dummy variable for whether the PCP and specialist are part of the same practice; and “Peer Outcomes” denotes the estimates of α when also including a measure of specialist quality based on the PCP’s peers (e.g., another measure of quality based solely on the outcomes of other PCPs’ patients in the same practice). All specifications are otherwise identical, where we also impose time-invariant congestion for computational ease.

Figure 1: Event Studies by Failure Cohort^a

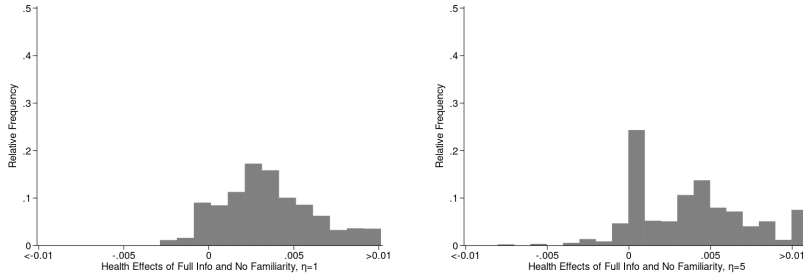


^aFigure depicts event study estimates following Cengiz et al. (2019), based on Equation 1 of the main text, estimated separately by failure cohort (e.g., first failure, second failure, etc.). Our specification includes specialist X quarter fixed effects as well as an indicator for PCP ‘type’. The event study coefficient at time $\tau = -1$ is normalized to 0.

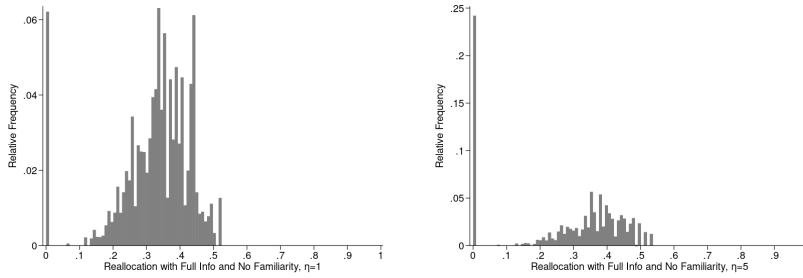
Figure 2: Counterfactuals from Full Information and No Familiarity^a
Myopic Referrals, Reallocation



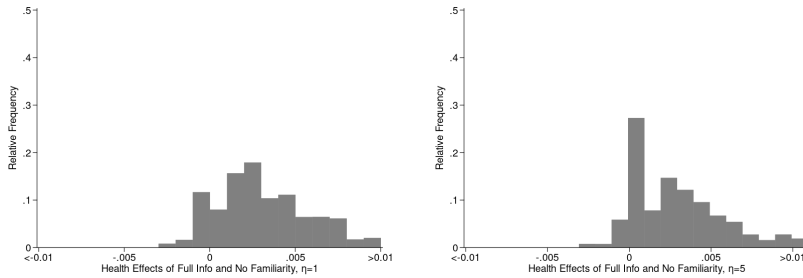
Myopic Referrals, Health Improvements



Forward-looking Referrals, Reallocation



Forward-looking Referrals, Health Improvements



^aFigure depicts the share of referrals reallocated and the resulting effects on patient health (i.e., the change in the probability of a successful surgery) under a counterfactual in which PCPs have full information about specialist quality and do not incorporate any familiarity in their referral decisions. Estimates are averaged across individuals within an HRR and presented as a histogram across HRRs. Results limited to HRRs in which our constrained maximization routine converged, 280 (277) HRRs in the myopic (forward-looking) case for $\eta = 1$ and 281 (279) in the myopic (forward-looking) case for $\eta = 5$.